Dirichlet Convolutions

Steven Breger

1 Arithmetic Functions

In all that follows, $\mathbb{N}$ denotes the set of positive integers.

Definition 1.1. *A function f is said to be arithmetic if f maps $\mathbb{N} \to \mathbb{C}$.*

Remark 1.2. We will mainly deal with functions that map $\mathbb{N} \to \mathbb{Z}$ in these notes.

Definition 1.3. *An arithmetic function f is said to be multiplicative if $f(mn) = f(m)f(n)$ for all coprime $m, n \in \mathbb{N}$ and if $f(1) \neq 0$.*

Definition 1.4. *An arithmetic function f is said to be completely multiplicative if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$ and if $f(1) \neq 0$.*

Note that for multiplicative or completely multiplicative f, $f(1) = f(1 \cdot 1) = f(1)^2$. Since $f(1) \neq 0$, $f(1) = 1$ (if we were to allow $f(1) = 0$, we would introduce the function $f(n) = 0$, which we will come to see does not play nicely with what follows). Consider σ, the sum of divisors function. We can write σ as

$$\sum_{d|n} d.$$

Let $D(n)$ be the set of divisors of n. It is not too hard to see that σ is multiplicative since there is a bijection between $D(mn)$ and $D(m) \times D(n)$ for coprime $m, n \in \mathbb{N}$. For instance, if we take $12 = 3 \cdot 2^2$, any divisor of 12 must have a component that comes from 3 and a component that comes from 2^2 (for example 6 can be split into 3 and 2 while 2^2 can be split into 1 and 2^2).

So, we can notice that if we write the sum of the divisors of 3: $\sigma(3) = 1 + 3$, the sum of the divisors of 2^2: $\sigma(2^2) = 1 + 2 + 2^2$, and the sum of the divisors of $3 \cdot 2^2$: $\sigma(3 \cdot 2^2) = 1 + 2 + 2^2 + 3 + 3 \cdot 2 + 3 \cdot 2^2$, the third is the product of the first, as when multiplying $\sigma(3)$ and $\sigma(2^2)$, each resulting term is exactly a divisor of $\sigma(3 \cdot 2^2)$ by our bijection. In general, we have that σ is multiplicative.

Because σ is multiplicative, if we can evaluate $\sigma(p^k)$ for all primes p and $k \geq 0$, we can evaluate $\sigma(n)$ for any n by prime factorizing n. If $p_1^{a_1} \cdots p_i^{a_i}$ is the canonical prime factorization of n, then $\sigma(n) = \sigma(p_1^{a_1} \cdots p_i^{a_i}) = \sigma(p_1^{a_1}) \cdots \sigma(p_i^{a_i})$.

2 The Dirichlet Convolution

Note the form we found for σ. We have that

$$\sum_{d|n} d.$$

We can generalize this by consider arithmetic functions F and f such that

$$F(n) = \sum_{d|n} f(d).$$

If you play around a little bit more, you will come to find that the following sum for 2 arithmetic functions f and g is very useful:

Definition 2.1. *(Dirichlet Convolution) Given arithmetic functions f and g, we define their Dirichlet Convolution to be $f * g$ where*

$$(f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right)$$

Example 2.2. $1 * \mathrm{id} = \sigma$. Note that here 1 refers to function that sends every natural to 1 and id refers to the identity function that sends every n to n. In other words, for all n, $1(n) = 1$ and $\mathrm{id}(n) = n$. Whether or not 1 refers to a number or a function should be clear based on context. The same is true for n.

Theorem 2.3. *Dirichlet convolution is commutative, associative, and distributes over addition.*

Proof. Commutativity and the distributive property are trivial. Note that to prove associativity, that instead of summing over $d \mid n$, we can some over all $ij = n$ and i will sum range over the divisors of n. So,

$$\begin{aligned}
&((f * g) * h)(n) \\
&= \sum_{kc=n} (f * g)(k)h(c) \\
&= \sum_{kc=n} \left(\sum_{ab=k} f(a)f(b) \right) h(c) \\
&= \sum_{kc=n} \sum_{ab=k} f(a)f(b)f(c) = \sum_{abc=n} f(a)f(b)f(c)
\end{aligned}$$

Analogously, the last quantity above is also equal to $f * (g * h)(n)$. This finishes. □

Theorem 2.4. *If any two of f, g, $f * g$ are multiplicative, then so is the third.*

Proof. Suppose that f and g are multiplicative. Let $m, n \in \mathbb{N}$ and coprime. Then, by the bijection between $D(mn)$ and $D(m) \times D(n)$ that we mentioned before,

$$\begin{aligned}
&(f * g)(mn) \\
&= \sum_{d|mn} f(d)g\left(\frac{mn}{d}\right) \\
&= \sum_{d_1|m} \sum_{d_2|n} f(d_1 d_2)g\left(\frac{mn}{d_1 d_2}\right) \\
&= \sum_{d_1|m} \sum_{d_2|n} f(d_1)f(d_2)g\left(\frac{m}{d_1}\right) g\left(\frac{n}{d_2}\right) \\
&= \left(\sum_{d_1|m} f(d_1)g\left(\frac{m}{d_1}\right) \right) \left(\sum_{d_2|n} f(d_2)g\left(\frac{n}{d_2}\right) \right) \\
&= (f * g)(m) \cdot (f * g)(n)
\end{aligned}$$

We now need to show that if g and $f * g$ are multiplicative, then so is f. Suppose the contrary. Then there exist $m, n \in \mathbb{N}$ such that $f(mn) \neq f(m)f(n)$. By the well ordering principle, we can choose m, n such

that mn is minimal. Consider that

$$\begin{aligned}
(f*g)(mn) &= \sum_{d|mn} f(d)g\left(\frac{mn}{d}\right) \\
&= f(mn) + \sum_{d|mn, d<mn} f(d)g\left(\frac{mn}{d}\right) \\
&= f(mn) + \sum_{d_1|m, d_1<m} \sum_{d_2|n, d_2<n} f(d_1d_2)g\left(\frac{mn}{d_1d_2}\right) \\
&= f(mn) + \sum_{d_1|m, d_1<m} \sum_{d_2|n, d_2<n} f(d_1)f(d_2)g\left(\frac{m}{d_1}\right)g\left(\frac{n}{d_2}\right) \quad \text{(by the minimality of } mn\text{)} \\
&= f(mn) + \sum_{d_1|m, d_1} \sum_{d_2|n, d_2} f(d_1)f(d_2)g\left(\frac{m}{d_1}\right)g\left(\frac{n}{d_2}\right) - f(m)f(n) \\
&= f(mn) + \sum_{d_1|m} f(d_1)g\left(\frac{m}{d_1}\right) \sum_{d_2|n} f(d_2)g\left(\frac{n}{d_2}\right) - f(m)f(n) \\
&= f(mn) + (f*g)(m)\cdot(f*g)(n) - f(m)f(n)
\end{aligned}$$

Since $f*g$ is multiplicative, the above implies that $f(mn) = f(m)f(n)$, a contradiction. □

Remark 2.5. As we will see, this theorem becomes false if we replace "multiplicative" with "completely multiplicative."

3 More about $*$

One natural question to ask about convolution is whether or not there is an arithmetic function ε that acts as an identity under convolution. That is for any arithmetic function f, $f*\varepsilon = f$. It is not too hard to see that

$$\varepsilon(n) = \begin{cases} 1, n = 1 \\ 0, \text{otherwise} \end{cases}$$

satisfies this property. It is up to the reader to verify this.

Remark 3.1. This is not the identity function on the naturals (the one denoted as id or n that sends every natural to itself)! id is the identity function on the naturals, which is not an identity under convolution as $f*\text{id}$ will not give you f most of the time! When we refer to an identity, the context will make it clear which function is being referenced. As far as math notation though, we will use ε if and only if we are referring to the identity under convolution.

Another natural question to ask is whether every arithmetic function f has a Dirichlet inverse. Well, suppose we have a function f^{-1} satisfying $f*f^{-1} = \varepsilon$. By plugging in 1 on both sides, we get that $f^{-1}(1) = \frac{1}{f(1)}$. Thus, $f(1) \neq 0$. Now, for any $n > 1$ we have that

$$\begin{aligned}
&(f*f^{-1})(n) = \varepsilon(n) \\
\implies &\sum_{d|n} f^{-1}(d)f\left(\frac{n}{d}\right) = 0 \\
\implies &f^{-1}(n)f(1) + \sum_{d|n, d<n} f^{-1}(d)f\left(\frac{n}{d}\right) = 0 \\
\implies &f^{-1}(n) = -\frac{1}{f(1)} \sum_{d|n, d<n} f^{-1}(d)f\left(\frac{n}{d}\right)
\end{aligned}$$

This gives us a recursive way of computing f^{-1} as long as $f(1) \neq 0$. So to summarize, there is an identity under $*$ and every function with $f(1) \neq 0$ has an inverse. Note that by definition, we get that each multiplicative (and completely multiplicative) function has an inverse.

What's cool is that with all of these properties, we get that arithmetic functions form a ring under $+$ and $*$ (note that they don't form a field because not every nonzero element has an inverse under $*$ by the above). Essentially what this means is that multiplication and addition are nice, but not everything has an inverse under $*$. This is analogous to how $\mathbb{Z}$ functions under $+$ and $\cdot$.

Also, the set of arithmetic functions with $f(1) \neq 0$ forms the group of units of this ring. This is another way of saying "the set of elements with an inverse" (think about $\{1, -1\}$ for $\mathbb{Z}$). The reason we say group is because this set forms a group under $*$, meaning its closed under $*$ and satisfies a few nice properties. Furthermore, the set of multiplicative arithmetic functions is a subgroup of this group of units (mainly because it satisfies closure under $*$ and inverses). The set of completely multiplicative functions is not though, since it is not closed under $*$, simply consider $1 * 1$.

It's not to important exactly what the necessary properties are (you can look them up if you like) but they are mostly the things proven above.

4 Möbius Inversion

Although we now know that we can theoretically compute the inverse of any multiplicative function, let us try to explicitly compute the inverse of the function 1.

Let $1^{-1} = \mu$. For starters, μ must be multiplicative by theorem 2.4. Thus, $\mu(1) = 1$. Furthermore, if we can determine the value of μ on each prime power we would be done since if we decompose any integer n into its prime canonical factorization $p_1^{a_1} \cdots p_i^{a_i}$, then we can compute $\mu(n)$ by using

$$\mu(n) = \mu(p_1^{a_1} \cdots p_i^{a_i}) = \mu(p_1^{a_1}) \cdots \mu(p_i^{a_i}).$$

Well, for any prime p and integer $a > 0$ we must have that

$$\begin{aligned} &(\mu * 1)(p^a) = \varepsilon(p^a) \\ \implies &\sum_{d|p^a} \mu(d) = 0. \end{aligned}$$

We can start by setting $a = 1$. We get that $\mu(p) = -1$. Then, by setting $a = 2$ we get that $\mu(p^2) = 0$. It is not too hard to see that in fact $\mu(p^a) = 0$ for all $a > 1$. In other words, μ zeros out when there is a prime of multiplicity more than 1 and gives -1 for a prime otherwise.

By using the prime factorization above, we can come up with the following

Definition 4.1. *We define the arithmetic function μ as follows.*

$$\mu(n) = \begin{cases} 1, n = 1 \\ (-1)^i, n = p_1 \cdots p_i \\ 0, \text{otherwise} \end{cases}$$

Of course, p_i are distinct primes in the above, and $\mu(n) = 0$ is equivalent to $p^2 \mid n$ for some prime p.

We have determined that $\mu = 1^{-1}$. This lets us do a cool trick.

Theorem 4.2. *(Möbius Inversion) Suppose that we have an arithmetic function f. Then,*

$$F(n) = \sum_{d|n} f(d).$$

if and only if

$$f(n) = \sum_{d|n} F(d)\mu\left(\frac{n}{d}\right).$$

In other words, $F = f * 1$ if and only if $f = F * \mu$.

Proof. Of course, $F = f * 1$. Because convolution is associative, convoluting both sides with μ gives us

$$\begin{aligned} & F * \mu = f * 1 * \mu \\ \iff & F * \mu = f * \varepsilon \\ \iff & F * \mu = f \end{aligned}$$

This finishes. □

Example 4.3. For a prime p and $a > 0$, $\varphi(p^a) = p^a - p^{a-1}$. Observe that this formula is simply $(\mathrm{id} * \mu)(p^a)$. Since φ is multiplicative, or because id and μ are multiplicative, we have that $\varphi = \mathrm{id} * \mu$. Thus, $\mathrm{id} = \varphi * 1$ and we get that

$$n = \sum_{d|n} \varphi(d).$$

5 Sample (Easy) Problems

- Find $\mu(30)$.
- True or False: A completely multiplicative function has an inverse.
- True or False: Completely multiplicative functions are closed under convolution.
- True or False: The convolution of two multiplicative functions is multiplicative.
- Compute $(\varphi * \sigma)(6^5)$.

6 Sample (Hard) Problems

- Find

$$f(n) = \sum_{k=1}^{n} \mu(k) \left\lfloor \frac{n}{k} \right\rfloor.$$

- Show that

$$\sum_{k=1}^{n} \varphi(k) = \frac{1}{2}\left(\sum_{k=1}^{n} \mu(k) \left\lfloor \frac{n}{k} \right\rfloor^2 + 1\right).$$

- Show that

$$\sum_{1 \le k \le n, (k,n)=1} e^{2\pi i \cdot \frac{k}{n}} = \mu(n)$$

(Hint: Consider the summing over all $k \le n$ and inserting a function that is 0 unless $(k, n) = 1$). Generalize this to find

$$\sum_{1 \le k \le n, (k,n)=1} f\left(\frac{k}{n}\right)$$

in terms of

$$\sum_{1 \le k \le n} f\left(\frac{k}{n}\right).$$

- Do there exist zero divisors in the Dirichlet Ring? If yes, give and justify an example. If not, provide a proof of the fact. Note that a zero divisor in the Dirichlet Ring is an arithmetic function f different from $\mathbf{0}$ such that there exists an arithmetic g different from $\mathbf{0}$ satisfying $f * g = \mathbf{0}$.